

Cryptology Theory Assignment - 2

Encrypt / Decrypt using the Pen & Paper method.

- 1] Encrypt "CHASE YOUR DREAMS" using
 a. Caesar Cipher;
 b. Shift Cipher with Key 7; C. Rot 13

Solution:

a] Caesar Cipher.

$$C = E(P, K)$$

$$C = E(P + n) \mod 26$$

$n = 3$

$$C = E(P + n) \mod 26$$

$$C = (2+3) = 5 = F$$

$$H = (7+3) = 10 = K$$

$$A = (0+3) = 3 = D$$

$$S = (18+3) = 21 = V$$

$$E = (4+3) = 7 = H$$

$$Y = (24+3) = 27 \mod 26 = 1 = A$$

$$O = (14+3) = 17 = R$$

$$U = (20+3) = 23 = X$$

$$R = (17+3) = 20 = U$$

$$D = (3+3) = 6 \mod 26 = 6 = G$$

$$R = (17+3) = 20 \mod 26 = 20 = U$$

$$E = (4+3) = 7 \mod 26 = 7 = H$$

$$A = (0+3) = 3 \mod 26 = 3 = D$$

$$M = (12+3) = 15 \mod 26 = 15 = P$$

$$S = (18+3) = 21 \mod 26 = 21 = V$$

Plaintext = "CHASE YOUR DREAMS"

Ciphertext = "FKDVH ARXU GUH DPV"

b] Shift Cipher with Key 7.

$$C = E(P, K) \mod 26 = E(P + n)$$

$$C = E(P + 7) \mod 26$$

$$C = (2+7) = 9 \mod 26 = 9 \Rightarrow J$$

$$H = (7+7) = 14 \mod 26 = 14 \Rightarrow O$$

$$A = (0+7) = 7 \mod 26 = 7 \Rightarrow H$$

$$S = (18+7) = 25 \mod 26 = 25 \Rightarrow Z$$

$$E = (4+7) = 11 \mod 26 = 11 \Rightarrow L$$

$$Y = (24+7) = 31 \mod 26 = 5 = F$$

$$O = (14+7) = 21 \mod 26 = 21 = V$$

$$U = (20+7) = 27 \mod 26 = 1 = A$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$D = (3+7) = 10 \mod 26 = 10 = K$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$E = (4+7) = 11 \mod 26 = 11 = L$$

$$A = (0+7) = 7 \mod 26 = 7 = H$$

$$M = (12+7) = 19 \mod 26 = 19 = U$$

$$S = (18+7) = 25 \mod 26 = 25 = Z$$

$$C = (2+7) = 9 \mod 26 = 9 \Rightarrow J$$

$$H = (7+7) = 14 \mod 26 = 14 \Rightarrow O$$

$$A = (0+7) = 7 \mod 26 = 7 \Rightarrow H$$

$$S = (18+7) = 25 \mod 26 = 25 \Rightarrow Z$$

$$E = (4+7) = 11 \mod 26 = 11 \Rightarrow L$$

$$Y = (24+7) = 31 \mod 26 = 5 = F$$

$$O = (14+7) = 21 \mod 26 = 21 = V$$

$$U = (20+7) = 27 \mod 26 = 1 = A$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$D = (3+7) = 10 \mod 26 = 10 = K$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$E = (4+7) = 11 \mod 26 = 11 = L$$

$$A = (0+7) = 7 \mod 26 = 7 = H$$

$$M = (12+7) = 19 \mod 26 = 19 = U$$

$$S = (18+7) = 25 \mod 26 = 25 = Z$$

$$C = (2+7) = 9 \mod 26 = 9 \Rightarrow J$$

$$H = (7+7) = 14 \mod 26 = 14 \Rightarrow O$$

$$A = (0+7) = 7 \mod 26 = 7 \Rightarrow H$$

$$S = (18+7) = 25 \mod 26 = 25 \Rightarrow Z$$

$$E = (4+7) = 11 \mod 26 = 11 \Rightarrow L$$

$$Y = (24+7) = 31 \mod 26 = 5 = F$$

$$O = (14+7) = 21 \mod 26 = 21 = V$$

$$U = (20+7) = 27 \mod 26 = 1 = A$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$D = (3+7) = 10 \mod 26 = 10 = K$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$E = (4+7) = 11 \mod 26 = 11 = L$$

$$A = (0+7) = 7 \mod 26 = 7 = H$$

$$M = (12+7) = 19 \mod 26 = 19 = U$$

$$S = (18+7) = 25 \mod 26 = 25 = Z$$

$$C = (2+7) = 9 \mod 26 = 9 \Rightarrow J$$

$$H = (7+7) = 14 \mod 26 = 14 \Rightarrow O$$

$$A = (0+7) = 7 \mod 26 = 7 \Rightarrow H$$

$$S = (18+7) = 25 \mod 26 = 25 \Rightarrow Z$$

$$E = (4+7) = 11 \mod 26 = 11 \Rightarrow L$$

$$Y = (24+7) = 31 \mod 26 = 5 = F$$

$$O = (14+7) = 21 \mod 26 = 21 = V$$

$$U = (20+7) = 27 \mod 26 = 1 = A$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$D = (3+7) = 10 \mod 26 = 10 = K$$

$$R = (17+7) = 24 \mod 26 = 24 = Y$$

$$E = (4+7) = 11 \mod 26 = 11 = L$$

$$A = (0+7) = 7 \mod 26 = 7 = H$$

$$M = (12+7) = 19 \mod 26 = 19 = U$$

$$S = (18+7) = 25 \mod 26 = 25 = Z$$

$$C = (2+7) = 9 \mod 26 = 9 \Rightarrow J$$

$$H = (7+7) = 14 \mod 26 = 14 \Rightarrow O$$

$$A = (0+7) = 7 \mod 26 = 7 \Rightarrow H$$

$$S = (18+7) = 25 \mod 26 = 25 \Rightarrow Z$$

$$E = (4+7) = 11 \mod 26 = 11 \Rightarrow L$$

$$Y = (24+7) = 31 \mod 26 = 5 = F$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

Ciphertext

JOHYLFVBX KYLHUZ

CHASE YOUR DREAMS.

JOHYLFVBX KYLHUZ

c) ROT 13

ABCDEFGHIJKLMNOPQRSTUVWXYZ

$$C = (P + 13) \bmod 26$$

$$C = (2 + 13) \bmod 26 = 15 \bmod 26 = 15 \Rightarrow P$$

$$H = (7 + 13) \bmod 26 = 20 \bmod 26 = 20 \Rightarrow U$$

$$A = (0 + 13) \bmod 26 = 13 \bmod 26 = 13 \Rightarrow N$$

$$S = (18 + 13) \bmod 26 = 31 \bmod 26 = 5 \Rightarrow F$$

$$E = (4 + 13) \bmod 26 = 17 \bmod 26 = 17 \Rightarrow R$$

$$Y = (24 + 13) \bmod 26 = 37 \bmod 26 = 11 \Rightarrow L$$

$$O = (14 + 13) \bmod 26 = 27 \bmod 26 = 1 \Rightarrow B$$

$$U = (20 + 13) \bmod 26 = 33 \bmod 26 = 7 \Rightarrow H$$

$$R = (17 + 13) \bmod 26 = 30 \bmod 26 = 4 \Rightarrow E$$

$$P = (15 + 13) \bmod 26 = 28 \bmod 26 = 2 \Rightarrow C$$

$$R = (17 + 13) \bmod 26 = 30 \bmod 26 = 4 \Rightarrow E$$

$$E = (4 + 13) \bmod 26 = 17 \bmod 26 = 17 \Rightarrow R$$

$$A = (5 + 13) \bmod 26 = 18 \bmod 26 = 18 \Rightarrow S$$

$$M = (12 + 13) \bmod 26 = 25 \bmod 26 = 25 \Rightarrow Z$$

$$S = (18 + 13) \bmod 26 = 31 \bmod 26 = 5 \Rightarrow F$$

PUNFR L B H E C E R S Z F

Ciphertext $\Rightarrow$

2. Decrypt:

(a) WRIND OOB VHEUHW PH VVDJH " using
caesar cipher.

~~Decrpt~~

$$P = D(C - n) \bmod 26 \quad n = 3$$

$$W = (22 - 3) = 19 \bmod 26 = T$$

$$R = (17 - 3) = 14 \bmod 26 = O$$

$$W = (22 - 3) = 19 \bmod 26 = T$$

$$D = (3 - 3) = 0 \bmod 26 = A$$

$$O = (14 - 3) = 11 \bmod 26 = L$$

$$O = (14 - 3) = 11 \bmod 26 = L$$

$$B = (1 - 3) = -2 \bmod 26 = 24 = Y$$

$$V = (21 - 3) = 18 \bmod 26 = S$$

$$H = (7 - 3) = 4 \bmod 26 = E$$

$$F = (5 - 3) = 2 \bmod 26 = C$$

$$U = (20 - 3) = 17 \bmod 26 = R$$

$$H = (7 - 3) = 4 \bmod 26 = E$$

$$W = (22 - 3) = 19 \bmod 26 = T$$

$$P = (15 - 3) = 12 \bmod 26 = M$$

$$H = (7 - 3) = 4 \bmod 26 = E$$

$$V = (21 - 3) = 18 \bmod 26 = S$$

$$V = (21 - 3) = 18 \bmod 26 = S$$

$$D = (3 - 3) = 0 \bmod 26 = A$$

$$J = (9 - 3) = 6 \bmod 26 = G$$

$$H = (7 - 3) = 4 \bmod 26 = E$$

Plain text: TOTALLY SECRET MESSAGE

(b) (1) MJQQT BTWQI using cipher 5

$$P = D(C - n) \quad n = 5$$

$$M = (12 - 5) \bmod 26 = 7 = H$$

$$J = (9 - 5) \bmod 26 = 4 = E$$

$$Q = (16 - 5) \bmod 26 = 11 = L$$

$$Q = (16 - 5) \bmod 26 = 11 = L$$

$$T = (19 - 5) \bmod 26 = 14 = O$$

$$B = (1 - 5) \bmod 26 = 26 = W$$

$$T = (19 - 5) \bmod 26 = 14 = O$$

$$U = (22 - 5) \bmod 26 = 17 = R$$

$$Q = (16 - 5) \bmod 26 = 11 = L$$

$$I = (8 - 5) \bmod 26 = 3 = D$$

Plain text: HELLO WORLD

(2) STB DTWP YNRJ

$$S \Rightarrow (18 - 5) \bmod 26 = 13 = N$$

$$T \Rightarrow (9 - 5) \bmod 26 = 4 = E$$

$$B \Rightarrow (1 - 5) \bmod 26 = 26 = W$$

$$D = (3 - 5) \bmod 26 = 24 = Y$$

$$T = (19 - 5) \bmod 26 = 14 = O$$

$$W = (22 - 5) \bmod 26 = 17 = R$$

$$P = (15 - 5) \bmod 26 = 10 = K$$

$$Y = (24 - 5) \bmod 26 = 19 = T$$

$$W = (13 - 5) \bmod 26 = 8 = I$$

$$R = (17 - 5) \bmod 26 = 12 = M$$

$$J = (9 - 5) \bmod 26 = 4 = E$$

Plaintext = NEW YORK TIME

G. FRAQ ZBER PBSSRR using ROT 13.

solution

~~P=DE~~

$$P = (C - 13) \bmod 26.$$

$$F \Rightarrow (5 - 13) \bmod 26 = -8 \bmod 26 = 18 = S.$$

$$R \Rightarrow (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E.$$

$$A \Rightarrow (0 - 13) \bmod 26 = -13 \bmod 26 = 13 = N.$$

$$Q \Rightarrow (16 - 13) \bmod 26 = 3 \bmod 26 = 3 = D.$$

$$Z \Rightarrow (25 - 13) \bmod 26 = 12 \bmod 26 = 12 = M.$$

$$B \Rightarrow (1 - 13) \bmod 26 = -12 \bmod 26 = 14 = O.$$

$$E \Rightarrow (4 - 13) \bmod 26 = -9 \bmod 26 = 17 = R.$$

$$R \Rightarrow (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E.$$

$$P \Rightarrow (15 - 13) \bmod 26 = 2 \bmod 26 = 2 = C.$$

$$B \Rightarrow (1 - 13) \bmod 26 = -12 \bmod 26 = 14 = O.$$

$$S \Rightarrow (18 - 13) \bmod 26 = 5 \bmod 26 = 5 = F.$$

$$S \Rightarrow (18 - 13) \bmod 26 = 5 \bmod 26 = 5 = F.$$

$$R \Rightarrow (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E.$$

$$R \Rightarrow (17 - 13) \bmod 26 = 4 \bmod 26 = 4 = E.$$

~~Plain~~ Plain text = SEND MORE COFFEE

Plain text = SEND MORE COFFEE.